

TermPlanMT: Graph-Grounded Terminology Planning for French–English Regulatory Medical Document Translation

When Ontology Grounding Helps and When It Does Not

Darragh Russell Kerins
UPV/EHU

drussell001@ikasle.ehu.eus

Noumidia Abadour
UPV/EHU

nabadour001@ikasle.ehu.eus

Abstract

Terminology consistency is a patient-safety requirement in regulatory medical translation, yet neural MT systems are structurally indifferent to cross-segment term stability. We present **TermPlanMT**, a five-system ablation study on Section 4.8 of the KEYTRUDA SmPC—a 126-segment French–English corpus with maximum adverse-event density. Our central finding is that Concept Coverage Rate (CCR), the fraction of source spans grounded in a MedDRA ontology graph, is the binding constraint on terminology-aware MT: at 35% CCR graph-free baselines (S1–S2) outperform all graph-augmented systems, while improving NER quality yields ≈ 7 BLEU gains for Mistral graph systems (S3–S5-Mistral) but not for graph-free S1 or S5-NLLB; S2 lies in between with a wide bootstrap interval (Table 5). We further show that our Hierarchy-Aware Terminology Metric (HTM) scores the certified human reference at about 0.13–0.18 depending on NER manifest, a score unchanged by full MedDRA ancestor alias expansion, confirming a structural mismatch between MedDRA nomenclature and professional translation practice. Finally, ambiguous MedDRA grounding causes the planning stage to abstain from locking critical terms, explaining a paradox in which graph-augmented systems produce *lower* cross-document consistency than graph-free baselines. We complement aggregate scores with segment-level qualitative analysis (missing-term rates, drift, modal agreement) exported under `error_analysis/` in the project repository.

1 Introduction

EMA Summaries of Product Characteristics (SmPCs) require a single adverse-event concept to receive one identical English rendering across an entire document—a requirement that affects pharmacovigilance signal detection and patient safety [?]. Neural MT handles medical prose fluently but is structurally indifferent to cross-segment consistency, producing term drift (same source term, multiple renderings), concept flattening (LLT-level terms collapsed to hypernyms), and coverage gaps (multi-token spans outside NER). MedDRA’s five-level hierarchy (Table 1) provides validated English labels that are an attractive grounding resource—if the grounding pipeline can cover enough of the source text.

Contributions.. (1) A five-system ablation ladder (§4) isolating the effects of NER recall, ontology grounding, and decoding enforcement. (2) HTM and alias-expanded HTM-A (§5), with a diagnostic evaluation showing the metric systematically penalises the human reference. (3) A CCR-bottleneck experiment (§6) showing that NER quality—not graph retrieval strategy—determines whether ontology grounding helps. (4) A mechanistic account (§8) of why graph-augmented systems are *less* consistent than graph-free ones. (5) Structured qualitative error analysis tying drift to ambiguous grounding and abstaining locks, with reproducible CSV/MD artefacts.

2 Related Work

Lexically constrained decoding [??] and terminology-aware training [?] enforce surface-level term appearance but ignore ontological hierarchy and do not address candidate selection when multiple levels are valid. ? show that extending NMT context windows can degrade fluency, moti-

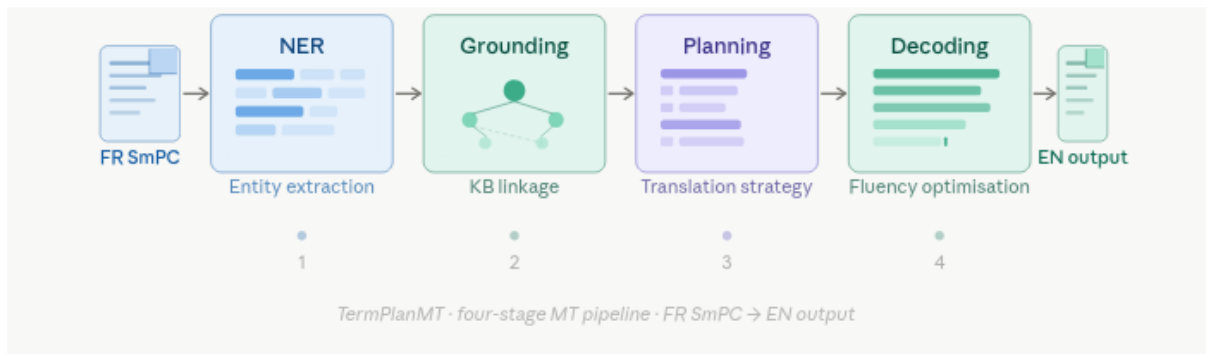


Figure 1: Four-stage TermPlanMT pipeline (NER → grounding → planning → decoding).

HTM — Hierarchy-Aware Terminology Metric

HTM measures how far a translated term has drifted from its source position in the MedDRA hierarchy. Each MT output term is grounded in Neo4j to recover its level $\hat{d}_i$, then compared against the source level d_i :

$$\text{HTM} = \frac{1}{N} \sum_{i=1}^N s_i, \quad s_i = \begin{cases} 1.0 & \hat{d}_i = d_i \text{ (exact level match)} \\ 0.5 & \hat{d}_i \neq d_i, \text{ same branch (wrong specificity)} \\ 0.0 & \text{different branch or ungrounded} \end{cases} \quad (2)$$

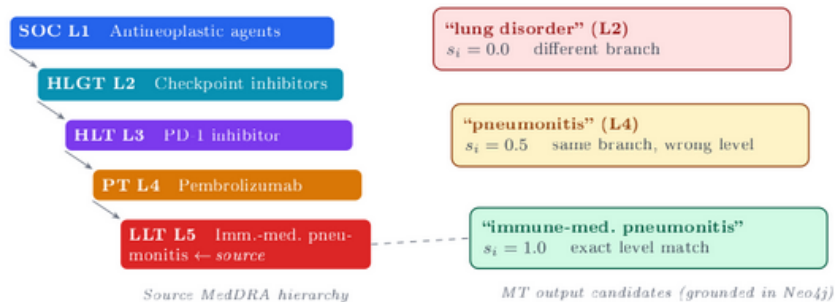


Figure 2: HTM scoring for *pneumopathie immuno-médicée* (source: LLT, L5). Each MT output is grounded in Neo4j and its MedDRA level compared against the source.

vating document-level LLMs as a consistency baseline. ? release BioMistral, the medically adapted LLM used for NER here. ? identify adverse-event terminology as a high-priority MT failure mode. GraphRAG-style retrieval [?] into LLM prompts provides the template for our S3 system, adapting knowledge-grounded MT [?] to a Neo4j MedDRA backend.

3 Dataset

Corpus.. Section 4.8 (Undesirable Effects) of the KEYTRUDA (pembrolizumab) SmPC [?]: 126 aligned French–English segment pairs after excluding segment 48_028 (a dense frequency table whose n-gram overlap distorts corpus metrics). Coverage: $\approx 4,400$ words; all three failure modes at high density.

Level	Abbr.	Example
L1	SOC	Respiratory disorders
L2	HLGT	Pulmonary conditions
L3	HLT	Pneumonitis
L4	PT	Pembro. pneumonitis
L5	LLT	Immune-med. pneumonitis

Table 1: MedDRA hierarchy. Adverse-event mentions in clinical text map almost exclusively to L5 ($\approx 96\%$).

NER conditions as a CCR perturbation..

Rather than comparing two NER systems, we use them as a controlled perturbation of CCR within the same pipeline. **Baseline NER** (ner_bio11m): BioMistral-7B zero-shot; CCR = 35.4%, 167 grounded spans. This condition exhibits a systematic artefact—the model generates *hypothyroïdie* for 92/127 segments regardless of

content—partially contaminating consistency measurements. **Fine-tuned (higher-quality) NER** (`ner_biol1m_ft`): BioMistral-7B fine-tuned on a hierarchical MedDRA SFT dataset; 170 grounded spans with cleaner boundaries and no hallucinated terms. Neo4j dataset CCR on the evaluation manifest (127 segments, incl. 48_028) is $\approx 36.4\%$. Graph-augmented systems benefit substantially under this condition, confirming that *effective* span quality (clean boundaries, usable grounding)—not raw graph retrieval—is the binding constraint once contamination is controlled.

4 System Architecture

Five systems form an ablation ladder (Table 2). All share the same four-stage pipeline: French NER $\rightarrow$ MedDRA grounding via Neo4j $\rightarrow$ global terminology lock $\rightarrow$ MT decoding.

ID	Model	Graph	Enforce.
S1	NLLB-200 600M	No	None
S2	Mistral-7B (doc)	No	None
S3	Mistral + GraphRAG	Yes	None
S4	S3 + reranking	Yes	Soft
S5	NLLB/Mistral + logit	Yes	Hard

Table 2: Ablation ladder. S1 = NMT lower bound; S2 = LLM with full document context; S3 is the key novel condition.

Terminology planning.. Before translation, `compute_global_locks()` scores each candidate English rendering c :

$$\text{Score}(c) = \alpha \text{Ctx}(c) + \beta \text{Nbr}(c) - \gamma \text{HP}(c) \quad (1)$$

where Ctx is cosine similarity to source segments containing the term, Nbr is similarity to adjacent MedDRA concept renderings, and $\text{HP} = |\hat{d} - d_{\text{src}}|$ penalises wrong specificity level ($\alpha=0.5$, $\beta=0.3$, $\gamma=0.2$). When a term matches *multiple* MedDRA concepts, no lock is committed (§8).

5 Metrics

Fluency.. Corpus BLEU [?] and chrF++ [?] via SacreBLEU against one professional English reference.

HTM and HTM-A.. For each segment containing a grounded French span, the hypothesis is scored against the MedDRA graph: $s_i = 1.0$ (exact level match), 0.5 (same branch, wrong specificity), or 0.0 (different branch / absent). $\text{HTM} = \text{mean}(s_i)$. HTM-A relaxes matching across three alias tiers: T1 exact LLT; T2 adds PT parent; T3 adds all same-branch ancestors. We compute **rHTM** by applying HTM to the professional reference as hypothesis, establishing a performance ceiling.

Hyp-Ref Agreement (HRA).. Fraction of HTM-measured segments where the system’s terminology choice matches the professional reference—a measure of human-like translation independent of MedDRA correctness.

CCR and drift.. $\text{CCR} = \text{grounded}/\text{extracted}$ spans; identical across S1–S5 for a given NER condition. Term drift = $(D - 1)/(N - 1)$ for D distinct renderings across N segments; drift = 0 is perfect consistency.

6 Results

6.1 Main Results

System	BLEU	chrF++	HTM	HRA
S1 NLLB	18.89	31.90	0.114	0.681
S2 Mistral	16.70	31.83	0.196	0.657
S3 GraphRAG	10.41	31.79	0.253	0.590
S4 rerank	10.10	32.01	0.237	0.596
S5 NLLB+boost	17.70	30.88	0.135	0.663
S5 Mist+boost	10.35	31.74	0.267	0.578
Reference	—	—	0.130	1.000

Table 3: Baseline NER condition (`ner_biol1m`, CCR = 35.4%; 127 segments). Corpus BLEU/chrF++ recomputed on stored JSONL (SacreBLEU); HTM/HRA from the latest full Neo4j eval when available. The human reference scores only 0.130 on HTM.

6.2 CCR is the Binding Constraint

Table 5 isolates the effect of NER quality by holding all other pipeline components fixed and comparing the same system across NER conditions.

System	BLEU	chrF++	HTM	HRA
S1 NLLB	18.89	31.90	0.195	0.435
S2 Mistral	20.37	35.85	0.274	0.497
S3 GraphRAG [†]	17.49	32.01	0.246	0.450
S4 rerank [†]	17.52	32.04	0.246	0.461
S5 NLLB+boost	17.54	31.02	0.205	0.408
S5 Mist+boost [†]	17.50	32.35	0.248	0.455
Reference	—	—	0.180	1.000

Table 4: Fine-tuned (higher-quality) NER condition (ner_biollm_ft; Neo4j dataset CCR $\approx 36.4\%$ on 127 segments incl. 48_028). [†]Contamination-filtered (18/127 rows per system scored as empty hypotheses; raw scores biased by prompt-injected MedDRA strings: BLEU deflated, chrF inflated).

System	Δ BLEU	Δ chrF++	95% CI Δ BLEU
S1 NLLB	± 0.00	± 0.00	[0.00, 0.00]
S2 Mistral	+3.67	+4.02	[−1.16, +7.95]
S3 GraphRAG	+7.07	+0.22	[+1.07, +11.10]
S4 rerank	+7.42	+0.03	[+1.18, +11.53]
S5 NLLB+boost	−0.16	+0.14	[−0.98, +0.66]
S5 Mist+boost	+7.16	+0.61	[+1.10, +11.66]

Table 5: Paired gain (fine-tuned minus baseline NER) on 127 segments with the same references. Δ chrF++ is the paired corpus difference. **95% bootstrap CIs** for Δ BLEU (2,000 resamples, seed 42): tools/eval/bootstrap_bleu_delta.py with -baseline-eval-file-set standard and -finetuned-eval-file-set mistral_clean (see tools/README.md). CIs for S3–S5-Mistral exclude 0; S2 and S5-NLLB intervals include 0.

S1 is unchanged (no graph). S2 shows a sizeable corpus Δ BLEU under better NER, but the paired bootstrap interval includes zero (Table 5). S3–S5-Mistral show ≈ 7 -point Δ BLEU with intervals strictly above zero, isolating the NER bottleneck on the graph path. S5-NLLB is effectively flat. Under fine-tuned NER, chrF++ rises for S2 and for Mistral graph runs, so the earlier “chrF drops when the graph is on” pattern is mostly a baseline-fluency artefact relative to the updated Mistral baseline JSONL.

Graph-free systems (S1–S2) remain above all graph-augmented rows on corpus BLEU/chrF++ at 35% CCR (Table 3) because the graph influences fewer than one third of adverse-event spans while injected MedDRA strings still hurt fluency where the graph is active. The contrast between near-zero Δ BLEU for S1/S5-NLLB, an uncertain S2 shift, and large Δ BLEU for S3–S5-Mistral in

Table 5 isolates the bottleneck upstream of decoding on the graph-augmented branch. Higher span coverage with cleaner NER would likely reverse the S2/S3 ranking—achievable with better French biomedical NER.

6.3 Missing Terms, Flattening, and Consistency

S5-Mistral achieves the lowest missing-term rate in both conditions (baseline: 29.5%; fine-tuned: 18.2%). S1-NLLB misses 69.88% (116/166) of grounded terms under baseline NER. Graph systems benefit from cleaner NER: S3’s missing-term rate falls from 33.1% to 19.4% (55/166 vs. 33/170 grounded spans). Mean cross-document modal agreement, averaged over French terms with ≥ 2 segments in error_analysis/*_consistency.csv, is highest for graph-free systems under baseline NER (S1: 0.993; S2: 0.984) while S3 scores 0.946; under fine-tuned NER, S3 rises to 0.976, surpassing S1 (0.970) but remaining below S2 (0.985)—so the “graph hurts consistency” pattern is clearest under noisy baseline NER and partially attenuates when NER improves, matching the missing-term and BLEU deltas.

Zero concept flattening events are observed: every grounded L5 source term produces an L5 hypothesis across all systems. This is partly an evaluation artefact—all gold terms are L5, and HTM cannot detect flattening unless upper-level aliases are provided—a limitation HTM-A addresses.

7 The HTM–Practice Gap

7.1 Why the Reference Scores Near Zero

On the baseline NER manifest, rHTM = 0.130–0.158 on the professional reference (Table 6); under fine-tuned NER the reference HTM is 0.180 (Table 4) because the grounded span set differs. In both cases the score stays below several MT systems on HTM. Professional translators do not transcribe MedDRA LLT strings verbatim: the LLT *Immune-mediated pneumonitis* appears as *pneumonitis* in tabular contexts and as *immune-related pneumonitis* in descriptive contexts.

Tier	Accepts	Low	Hi
T1	Exact LLT	0.130	0.158
T2	T1 + PT parent	0.130	0.158
T3	T2 + branch ancestors	0.130	0.158

Table 6: rHTM-A on the professional reference under three alias tiers. All tiers yield identical scores.

7.2 Alias Expansion Does Not Close the Gap

The ancestor chain for *pneumopathie inflammatoire* contains *pneumonitis*, so the null T1→T3 delta is not a graph coverage gap. Professional translators use phrasings such as *immune-related pneumonitis* and *inflammatory lung disease* that lie entirely outside the MedDRA preferred-label tree at any level. The mismatch is structural: MedDRA labels are designed for pharmacovigilance coding, not regulatory prose.

HTM should therefore be interpreted as a measure of MedDRA-proximity, not translation quality. The HRA column is the complementary measure. Notably, graph-augmented systems score consistently *lower* on HRA than S2 (Table 3: S3 = 0.590 vs. S2 = 0.657; Table 4: S3 = 0.450 vs. S2 = 0.497): the graph pushes outputs toward MedDRA labels that professional translators avoid.

8 Consistency Failure Analysis

Term drift (qualitative).. *Pneumopathie inflammatoire* occurs in four baseline segments where it is grounded for S3/S4/S5-Mistral, and in eight fine-tuned segments (more NER hits for the same French string). Those systems emit two renderings—often *pneumonitis* in incidence-style sentences and *immune-mediated pneumonitis* elsewhere—yielding drift scores of 1/3 (baseline) and 1/7 (fine-tuned). S1 collapses to one rendering for this span in both conditions (drift 0). S2 is stable in baseline (drift 0) but shows non-zero drift under fine-tuned NER once the span appears in additional contexts. S3/S4/S5-Mistral thus exhibit the clearest cross-segment inconsistency for this exemplar.

Mean modal agreement (§6) shows the same pattern at corpus level: graph-free systems rank highest under baseline NER; graph-augmented systems recover partially under fine-tuned NER but do not overtake S2.

Mechanism.. *Pneumopathie inflammatoire* maps to two distinct MedDRA concepts in

the string cache (one of 7,679 ambiguous normalised keys). The planning stage sets `has_planning_lock = 0` for any ambiguous term and commits no lock. S3/S4/S5-Mistral then resolve the ambiguity context-locally per segment from RAG context: incidence-focused segments attract *pneumonitis*; mechanism-focused segments attract *immune-mediated pneumonitis*. S1-NLLB produces zero drift not because it reasons about terminology, but because its fixed vocabulary has no mechanism for context-dependent selection.

This failure is independent of CCR. Even with perfect NER recall, any term matching multiple MedDRA concepts will be left unlocked, producing potential drift in all Mistral-based graph systems. The fix is *soft ambiguity resolution*: commit the top-scoring candidate as a provisional lock flagged for human review, consistent with CAT-tool workflows.

Output contamination.. In the fine-tuned NER condition, 18/127 segments per system (14.2%) contain prompt contamination: Mistral reproduced MedDRA metadata strings and instruction markers from its own system prompt. These rows were replaced with a null hypothesis for metric computation. Contamination biases raw scores in opposite directions—deflating BLEU (precision destroyed by injected strings) while inflating chrF (character overlap increased)—confirmed by S3 raw vs. filtered: BLEU 15.36→17.49, chrF 34.60→32.01. S1 and S2 are unaffected as they receive no graph context in their prompts.

9 Discussion and Limitations

The results collectively show that the TermPlanMT pipeline is architecturally sound but CCR-limited. Graph-augmented systems improve substantially when supplied with cleaner, higher-coverage NER input. The absence of BLEU movement for S1 (and near-zero change for S5-NLLB), together with large Mistral-graph ΔBLEU in Table 5, is direct evidence that the bottleneck lies upstream of graph retrieval on the graph-augmented path.

The HRA–HTM inversion—graph systems score higher on HTM yet lower on HRA—reveals a real tension: optimising toward MedDRA formal labels diverges from how certified translators actually write. For regulatory MT in practice, HRA is the more meaningful target.

Limitations.. *Single document and reference.* All results derive from one SmPC section against one professional translation, limiting generalisability. *HTM ceiling.* rHTM is low in absolute terms (about 0.13–0.18 across NER manifests) and invariant to alias expansion within a manifest, so HTM is only interpretable relatively. *NER artefact.* The baseline NER condition contains hallucinated spans (*hypothyroïdie* in 92/127 segments). *Contamination.* Fine-tuned-condition Mistral-based graph systems exhibit 14.2% prompt contamination; filtered scores are used throughout.

10 Conclusion

TermPlanMT demonstrates that CCR is the binding constraint on graph-augmented regulatory MT: at low coverage a graph-free LLM baseline dominates, while improving NER quality yields ≈ 7 -point corpus BLEU gains for Mistral graph systems with bootstrap intervals excluding zero. Three broader findings emerge: MedDRA nomenclature and professional translation practice are structurally misaligned in ways that alias expansion cannot fix; ambiguous grounding produces predictable term drift through a mechanism fully explained by the planning stage’s abstain-on-ambiguity policy; and consistency metrics must control for system type, since NLLB’s apparent consistency is an artefact of its fixed vocabulary. Substantially higher CCR with cleaner span boundaries would likely be required to reverse the S2/S3 fluency ordering; quantifying that threshold is left to future work with broader French biomedical NER.

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